

John Lee

AI & Backend Engineer — Singapore

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SUMMARY

AI engineer based in Singapore specialising in production-grade LLM platforms, RAG pipelines, and intelligent automation. Experienced in enterprise AI infrastructure, Kubernetes-based deployments, and end-to-end media processing pipelines. Open to full-time remote roles in AI/ML engineering and backend systems.

SKILLS

AI / LLM	LLM Integration, RAG Pipelines, Whisper STT, Vector Search, Edge TTS	Languages	Python (Expert), TypeScript (Advanced), SQL, Bash
Backend / Infra	Node.js, PostgreSQL, Docker, Kubernetes, Redis	Frameworks	FastAPI, MCP Protocol, GitHub Actions, Drizzle ORM

EXPERIENCE

AI Systems Contributor Jan 2026 – Present
Open Source · [archestra-ai/archestra](https://github.com/archestra-ai/archestra)

- Analyzed 877-line task queue with 5 handler types in a production enterprise AI platform
- Identified and fixed critical bug in attempt-counting logic affecting task retry behavior
- Mapped full architecture: K8s MCP runtime, knowledge base pipelines, and cache systems
- Contributed cost tracking and LLM usage monitoring optimizations

AI Pipeline Developer Jan 2025 – Present
Independent Projects · Singapore

- Built fully automated YouTube Shorts pipeline: Whisper transcription → translation → subtitle → upload (zero API cost)
- Developed IndoDub AI — multi-language video dubbing with neural TTS and audio mixing
- Implemented RAG pipelines and vector search integrations for LLM-powered applications
- Deployed services via Docker and Kubernetes for scalable production environments

PROJECTS

archestra — Enterprise AI Platform Contribution github.com/archestra-ai/archestra
TypeScript · PostgreSQL · Kubernetes · MCP · AI/LLM

- Fixed production bug in task queue attempt-counting logic; mapped K8s MCP runtime and KB pipelines

YouTube Shorts Pipeline github.com/64johnlee/youtube-pipeline
Python · Whisper · FFmpeg · YouTube API · Google Translate

- End-to-end automated pipeline: transcription → translation → SRT → upload with duplicate detection. Zero API cost.

IndoDub AI — Video Dubbing System github.com/64johnlee/indo-dub
Python · edge-tts · FFmpeg · Audio Processing

- Chinese → Indonesian neural dubbing with background music preservation and SRT generation

EDUCATION & LEARNING

Self-Taught AI & Backend Engineer 2023 – Present

- Deep study of LLM systems, RAG architecture, and production AI infrastructure
- Continuous learning via DeepLearning.AI, fast.ai, and open source codebase analysis